Sahaj Bhadja

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Summary

Systems-oriented Software Engineer with 2+ years applying **C++**, **algorithms**, simulation, automated verification, and reproducible evaluation to distributed software, security research, and correctness-sensitive production workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Restructured **fault-tolerant asynchronous orchestration** with explicit job-state transitions and failure isolation, processing 1K+ tasks while tripling throughput under concurrent workloads and preventing cascading failures across dependent services.
- Built data integrations across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, and CI/CD-backed releases at 99%+ measured production reliability.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, California*

- Built a PyTorch **simulation pipeline** processing 100K+ trajectory logs, improving long-horizon planning 18% through reproducible feature transformations, benchmark automation, regression testing, and detailed behavioral failure analysis workflows.
- Developed a spatial-temporal encoder combining map geometry, historical trajectories, and agent context, improving closed-loop stability 15% across complex scenarios through controlled evaluation and systematic root-cause debugging workflows.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Engineered cross-platform software with automated VM builds and 20+ **unit and integration suites** using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments reliably.
- Optimized object-oriented API integrations and AWS data flows, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20% through reusable services and structured reviews.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellow, Quantum-Resistant Systems *Gandhinagar, India*

- Analyzed consensus mechanisms and designed a quantum-resistant blockchain architecture integrating five lattice-based cryptographic primitives, reducing centralized-control and quantum-risk exposure 25% across documented comparative benchmark simulations and evaluations.
- Co-authored a peer-reviewed IEEE paper evaluating cryptographic and consensus properties, translating security assumptions and experimental results into a technical framework presented at the Indian Institute of Science.

Skills

Languages & Fundamentals: C++, Python, Java, TypeScript, JavaScript, SQL, Bash, Data Structures, Algorithms, OOP
Systems: Operating Systems, Concurrency, Multithreading, Distributed Systems, Networking, Microservices, State Machines, Simulation, Linux
Verification & Quality: Unit Testing, Integration Testing, Regression Testing, Automated Validation, Benchmarking, Debugging, Failure Analysis, PyTest, Mockito, Jest
Research & Infrastructure: Lattice Cryptography, Consensus Algorithms, Post-Quantum Security, Docker, Jenkins, CI/CD, Git, AWS, Azure, PostgreSQL

Projects

A Detailed Exploration to Quantum Resistant Blockchain Technology Paper | *FALCON, DPoSB, Borda Count*

- Co-authored an IEEE framework combining FALCON lattice-based signatures with DPoSB consensus, node-behavior monitoring, and Borda Count selection, then evaluated comparative throughput and latency across 100–1,000 transaction workloads.

CPU Scheduling Simulator Live | *C++, React, Operating Systems*

- Implemented six preemptive and non-preemptive scheduling algorithms with React controls, real-time Gantt visualization, and dynamic latency metrics across 1K+ process combinations, applying systems fundamentals to interactive resource management.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Designed a schema-aware transformation pipeline generating optimized SQL with 82% accuracy, implementing validation and structured execution across relational databases while supporting cross-platform development through Docker and Maven.